Integrating Cognition into Migration Models

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Existing Migration Studies vs. Empirical Reality

Traditional Models

Human mobility is typically characterized in binary contrasting portrayals:

- **Homogenous economic actors:** Utility-maximizing individuals.
- **Passive victims:** Driven by external forces like conflict, violence, or natural disasters.

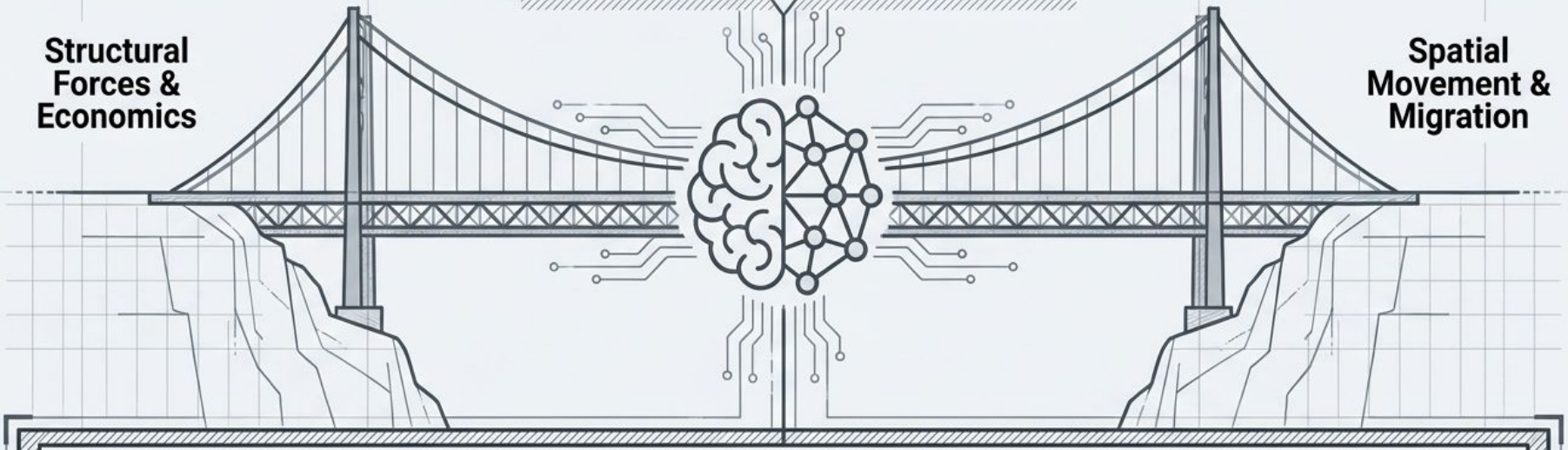
Empirical Reality

- Models fail to explain why individuals in **identical structural positions** make different migration decisions.
  - Some feel "trapped" between wanting to move and feeling unable to do so.
  - While others actively prefer to stay in home communities despite clear economic incentives to migrate.

To truly understand human mobility, it is imperative to move beyond rigid binaries. Migration is a multifaceted reality embedded in structural inequalities, cognitive landscapes, and social contexts.

Psychological factors such as attitudes, behavioral control, or social norms are seldom explicitly modeled.

— Klabunde and Willekens (2016)

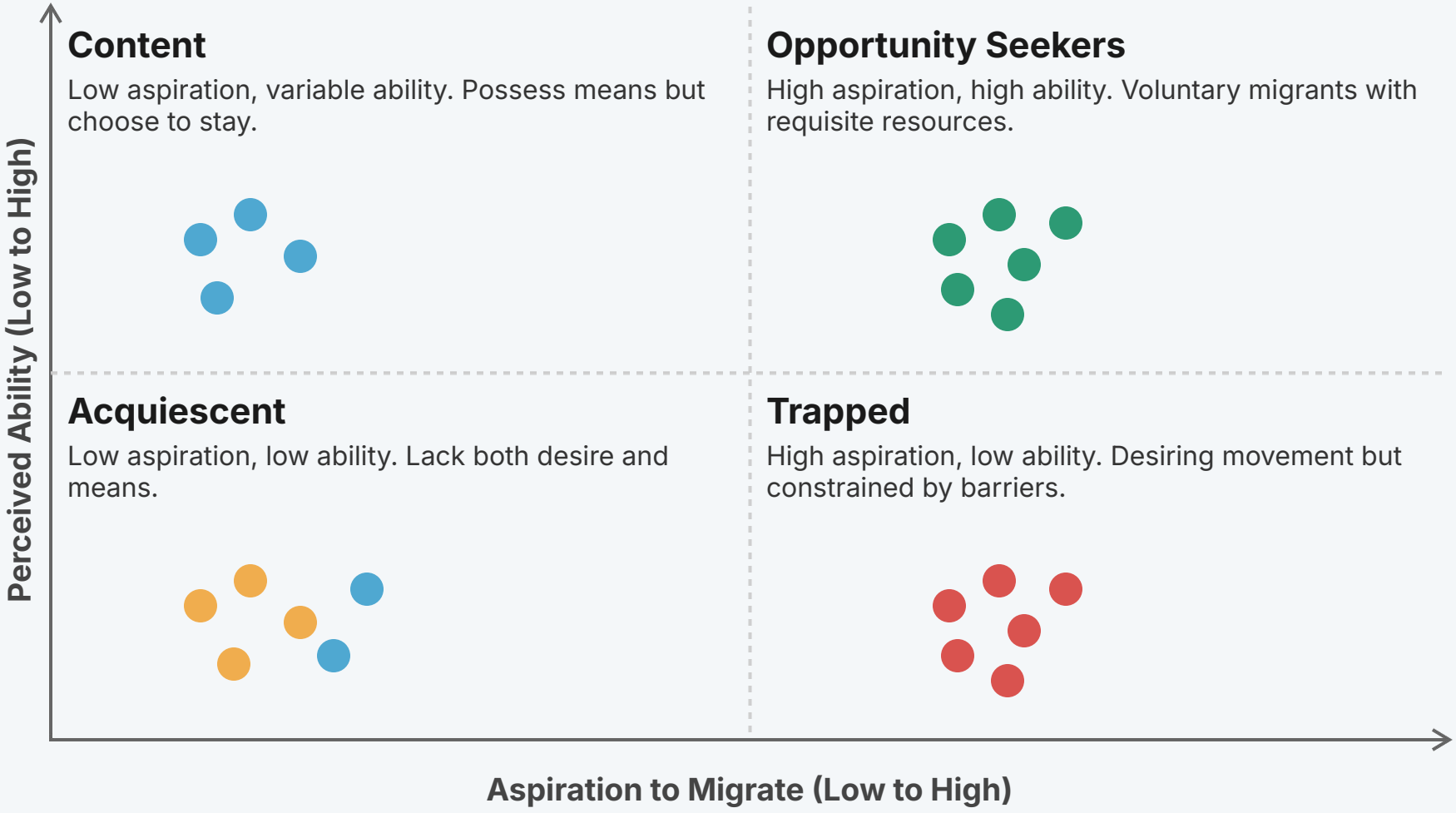


Carling & Schewel's (2018) **Aspiration-Ability Framework:**

Distinguishing the internal desire to migrate from the perceived external ability to do so.

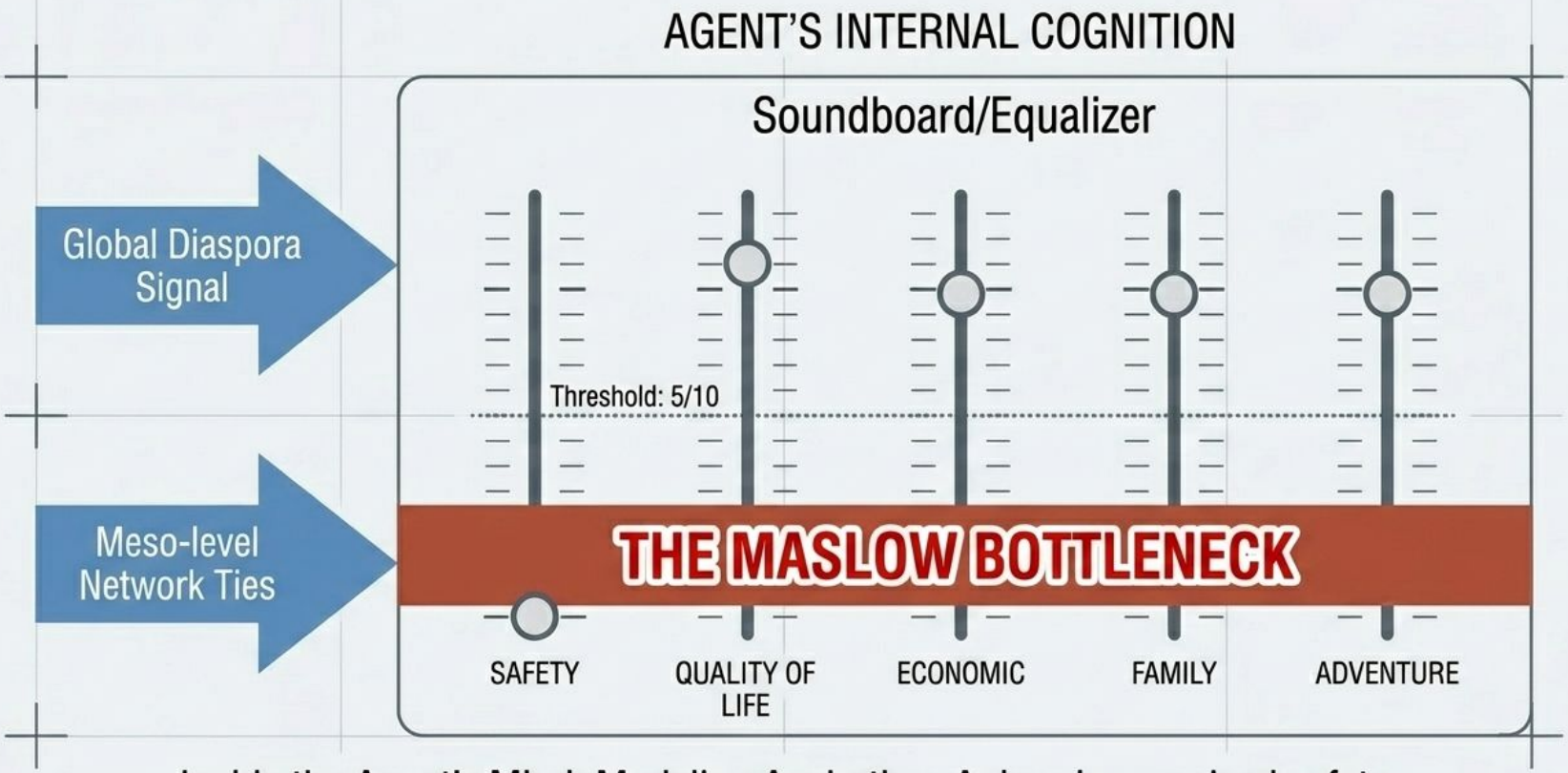
How can the theoretical aspiration-ability framework be operationalized into a computational Agent-Based Model (ABM)?

To introduce human agency and cognitive complexity into computational migration research, revealing the hidden, heterogeneous dynamics of global movement.

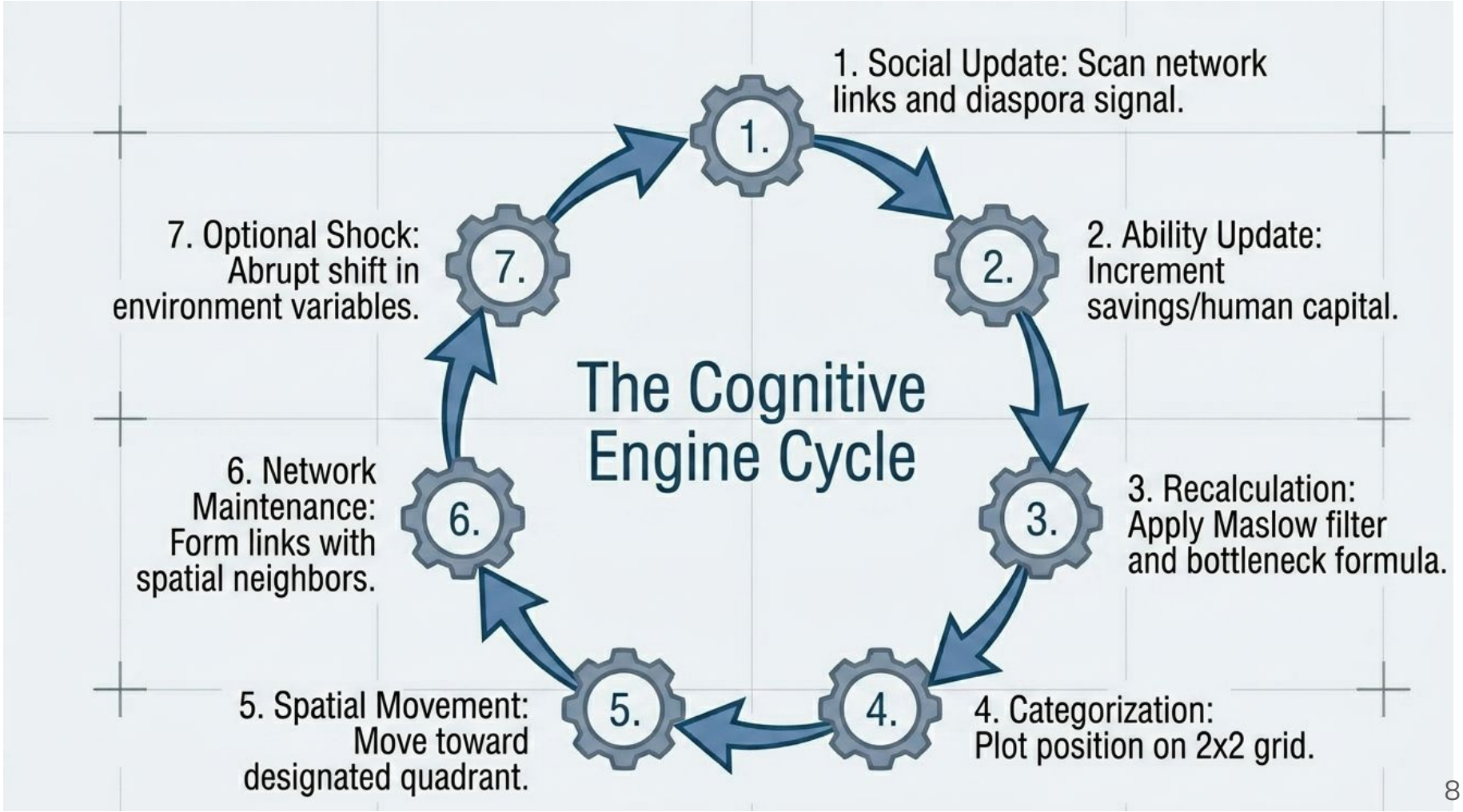

Content
Low aspiration, variable ability. Possess means but choose to stay.
Opportunity Seekers
High aspiration, high ability. Voluntary migrants with requisite resources.
Acquiescent
Low aspiration, low ability. Lack both desire and means.
Trapped
High aspiration, low ability. Desiring movement but constrained by barriers.
Perceived Ability (Low to High)
Aspiration to Migrate (Low to High)

Total Ability = (0.2 x min) + (0.8 x mean)

The Structural Reality: Modeling Ability. A compensatory system where significant bottlenecks—like a missing visa—heavily drag down the average, stranding an agent in the Low Ability threshold.



Inside the Agent's Mind: Modeling Aspiration. A drop in perceived safety halts aspirational growth, forcing a shift to immediate survival.


The Cognitive Engine Cycle
1.
1. Social Update: Scan network links and diaspora signal.
2.
2. Ability Update: Increment savings/human capital.
3.
3. Recalculation: Apply Maslow filter and bottleneck formula.
4.
4. Categorization: Plot position on 2x2 grid.
5.
5. Spatial Movement: Move toward designated quadrant.
6.
6. Network Maintenance: Form links with spatial neighbors.
7.
7. Optional Shock: Abrupt shift in environment variables.

Simulation Scenarios & Migration Phenomena

Run 1: Culture of Migration

Primary Driver
Social Networks/Human Capital

Aspiration State
Emerging/Evolving

Ability State
Steady Growth

Gap between Aspiration and Ability
Narrow & Manageable

Run 2: Economic Downturn

Primary Driver
External Disruption

Aspiration State
Spiked/Inflated

Ability State
Sharp Decline

Gap between Aspiration and Ability
Wide & Frustrated

Run 3: High Aspiration

Primary Driver
Capability Accumulation

Aspiration State
Constant (High)

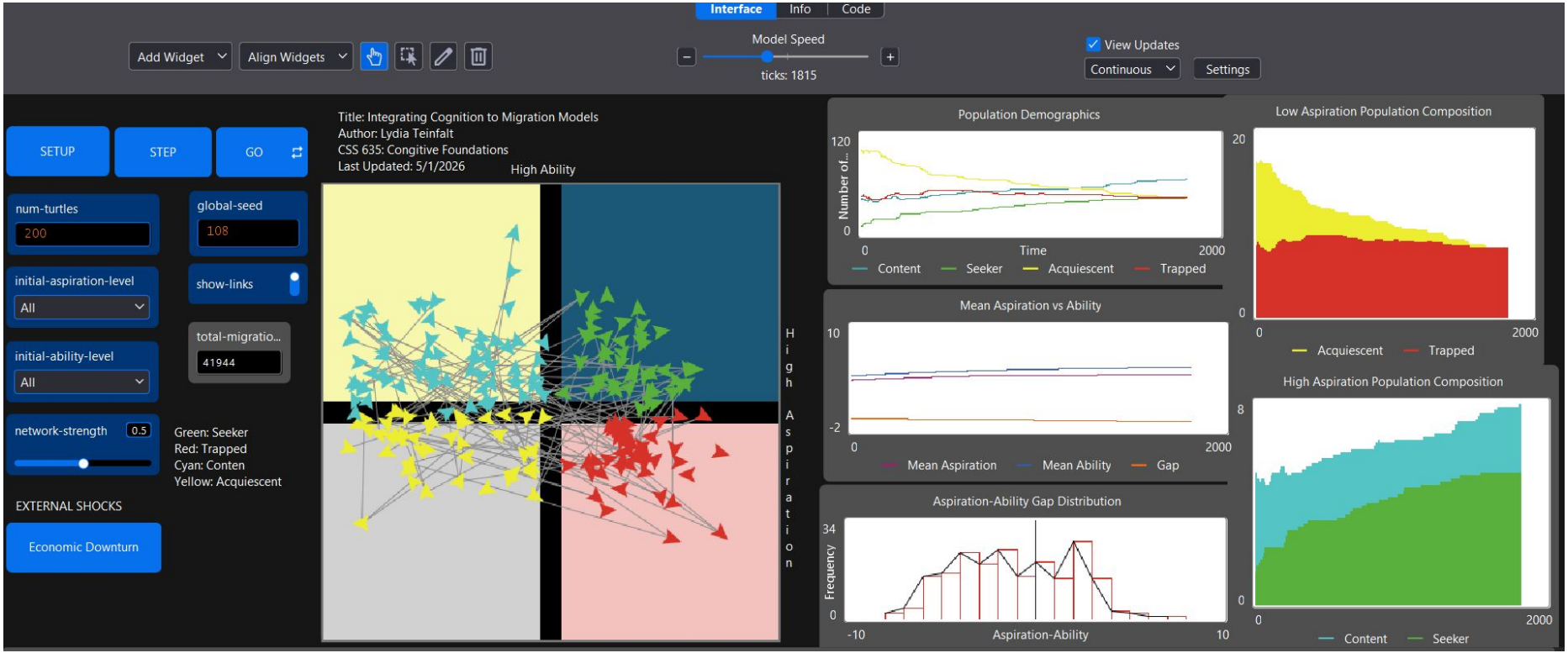
Ability State
Steady Growth

Gap between Aspiration and Ability
Closing over time

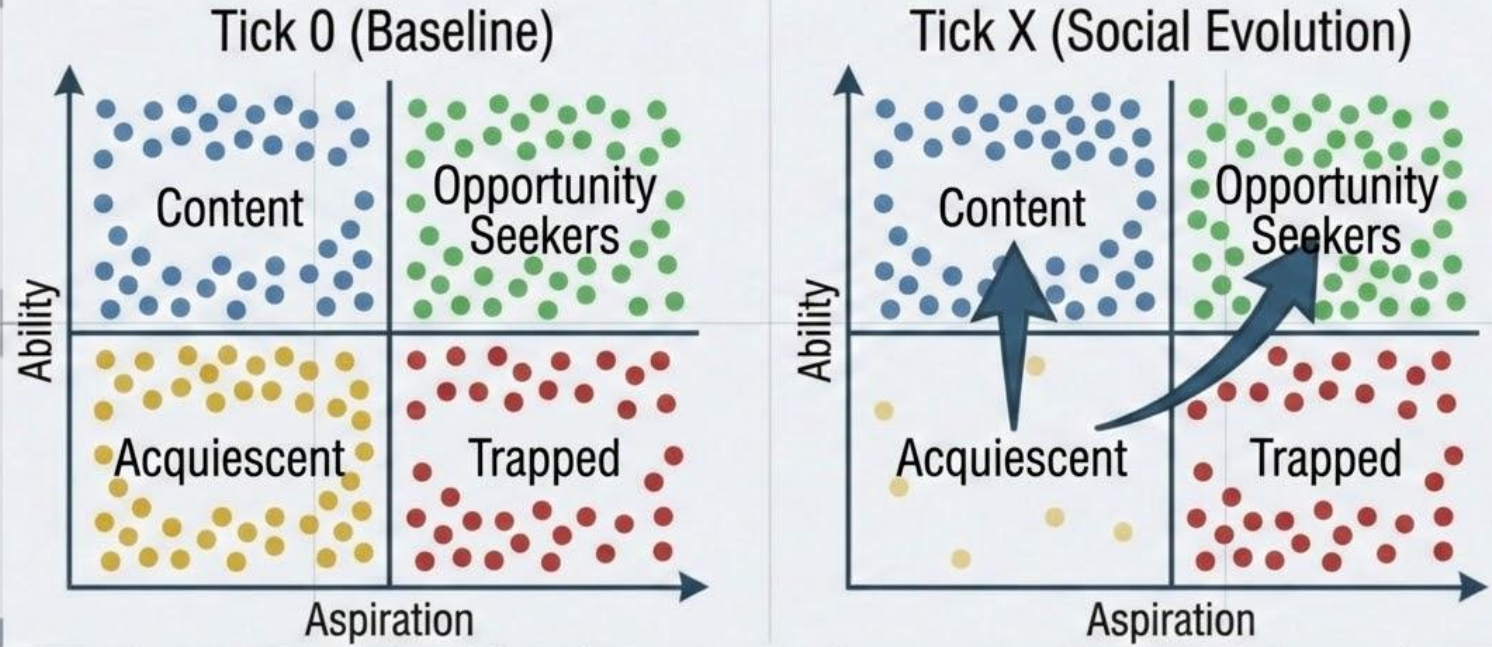
Model Scale: 200 Individual Agents

Simulation Duration: 1825 Ticks/Each tick = 1 day

NetLogo 7.03 Model

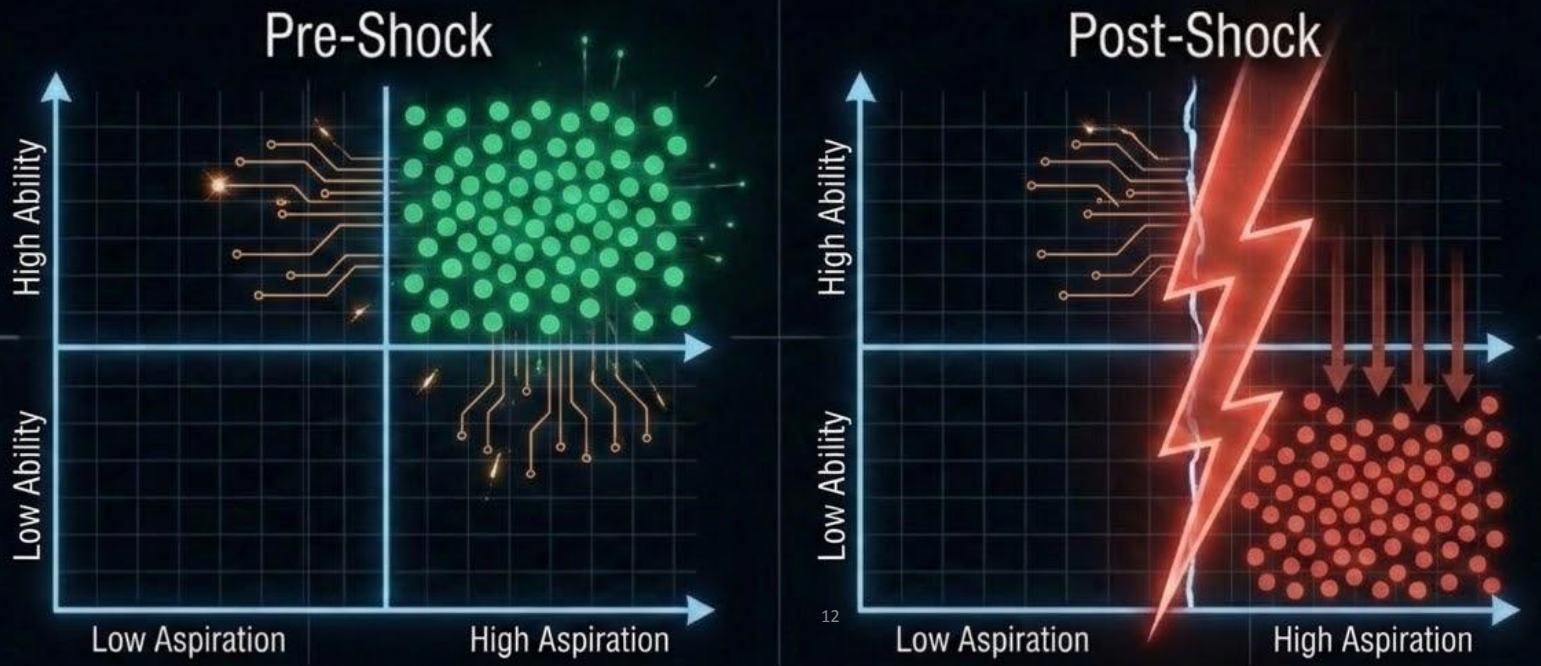


Baseline Simulation: The Culture of Migration



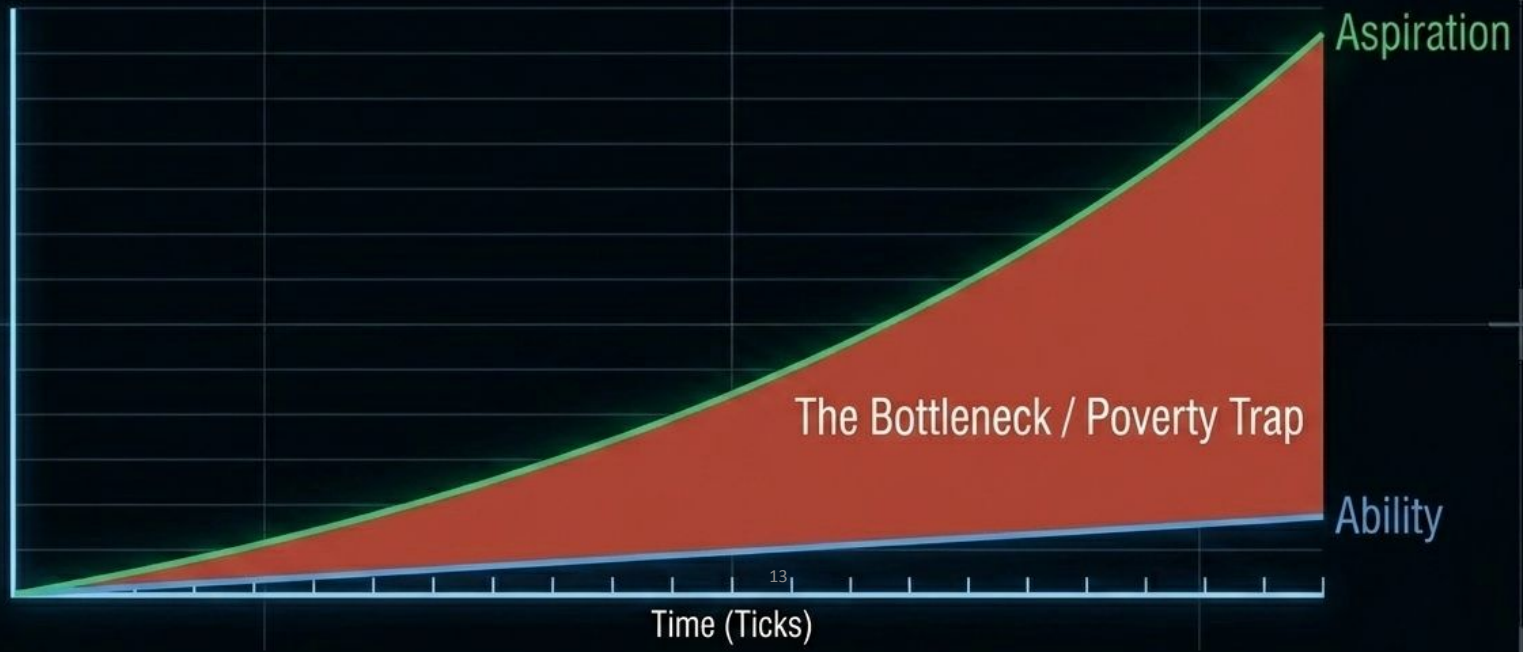
Insight: Natural social evolution. As the diaspora signal grows, the internal drive for those remaining increases. Aspiration is socially constructed and awakened by networks, regardless of immediate economic reality.

Simulation Shock: Creating Cognitive Misalignment



Insight: Systemic failures don't just stop movement—they strip away ability without erasing desire. Opportunity Seekers are instantly transformed into Trapped populations, creating a structural precursor to irregular migration.

Blocked Incremental Actualization: The Poverty Trap



Blocked Incremental Actualization. While ability grows linearly through savings, aspirations compound socially. The inescapable gap creates a permanent cognitive bottleneck for populations in precarious conditions.

MIGRATION IS NOT A SIMPLE RESPONSE TO POVERTY.

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Poverty is frequently a barrier to migration, not a driver. The most dangerous systemic state—the “Cognitive Gap”—occurs when aspirations remain high but structural ability is stripped away, leaving a frustrated population indefinitely trapped.

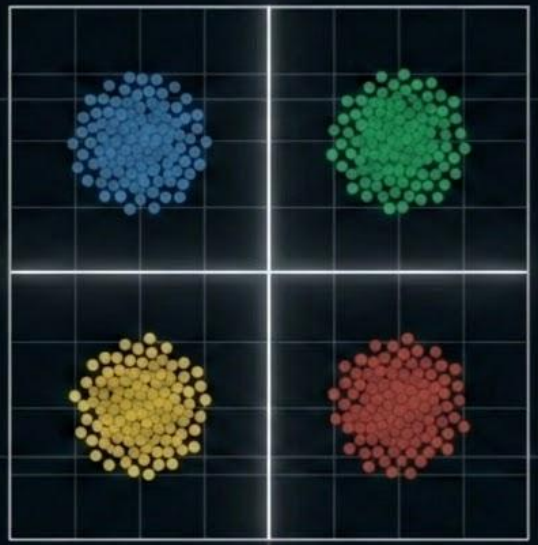
Validation Matrix: Theory to Observable ABM Result

Carling & Schewel (2018) Concept	Observable ABM Result
Involuntary Immobility (High desire blocked by walls)	Density of the Red Cluster
Social Construction of Aspiration (Desires shaped by networks)	Transition of Yellow nodes to Red/Green over time
Hierarchical Needs (Safety precedes migration)	Maslow Bottleneck logic halting total aspiration
Capability Accumulation (Migration requires capital thresholds)	Slow linear ability growth over time

Scope & Computational Limitations

The International Focus: Distance, information asymmetries, and hard structural barriers (like visas and language) uniquely magnify the gap between aspiration and ability.

Quantifying the Subjective: A fundamental limitation of Agent-Based Models is the necessary oversimplification of treating subjective human emotions as rigid Cartesian coordinates.



By integrating cognition, Agent-Based Models cease to be mere economic calculators. They become engines for empathy, capable of simulating the complex psychological state of global migration.

References

Alpes, M. J. 2014. “Imagining a Future in ‘Bush’: Migration Aspirations at Times of Crisis in Anglophone Cameroon.” Identities 21 (3): 259–274.
Carling, J., & Collins, F. (2018). Aspiration, desire and drivers of migration. Journal of Ethnic and Migration Studies, 44(6), 909–926. https://doi.org/10.1080/1369183X.2017.1384134
Carling, J., & Schewel, K. (2018). Revisiting aspiration and ability in international migration. Journal of Ethnic and Migration Studies, 44(6), 945–963. https://doi.org/10.1080/1369183X.2017.1384146
de Haas, H. A theory of migration: the aspirations-capabilities framework. CMS 9, 8 (2021). https://doi.org/10.1186/s40878-020-00210-4
Harris, J. R., & Todaro, M. P. (1970). Migration, Unemployment and Development: A Two-Sector Analysis. The American Economic Review, 60, 126-142.
Massey, D. S., Arango, J., Hugo, G., Kouaouci, A., Pellegrino, A., & Taylor, J. E. (1993). Theories of international migration: A review and appraisal. Population and Development Review, 19(3), 431-466. https://doi.org/10.2307/2938462
Maslow, A. H. (1943). A theory of human motivation. Psychological Review, 50(4), 370–396. https://doi.org/10.1037/h0054346

GitHub Repository

https://github.com/lydiateinfalt/Agent-Based-Modeling-Migration-and-Cognition
Selected images from slides were created with the NotebookLM and Advanced Gemini